

Input: QUBO matrix $Q \in \mathbb{R}^{n \times n}$; depth p ; device graph $G_{\text{dev}} = (V_{\text{dev}}, E_{\text{dev}})$ with two-qubit errors E_{uv} and readout errors r_u ; scale factors $\lambda_\gamma, \lambda_\beta$; shots S ; noise scales $\Lambda = \{1, 2\}$

Output: Mitigated energy $\hat{E}_0$; best bit string $\mathbf{x}^*$; QNRE score

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1  $s_i \leftarrow \sum_j |Q_{ij}|$ ;  $w_i \leftarrow s_i / \sum_l s_l$ ;  $w_{ij} \leftarrow |Q_{ij}| / \sum_{k < l} |Q_{kl}|$ ; // QPN
2 for  $k \leftarrow 1$  to  $p$  do
3    $\gamma_k^0 \leftarrow \frac{\lambda_\gamma k}{np} \sum_i \arccos(1 - 2w_i)$ ;  $\beta_k^0 \leftarrow \frac{\lambda_\beta (p-k)}{np} \sum_i \arcsin \sqrt{w_i}$ ; // direct init.
4    $\sigma \leftarrow$  logical qubits sorted by  $w_i$  in decreasing order;
5    $\bar{E}_u \leftarrow r_u + \sum_{v: (u,v) \in E_{\text{dev}}} E_{uv}$  for all  $u \in V_{\text{dev}}$ ; // aggregated error
6    $\pi(\sigma_1) \leftarrow \arg \min_{u \in V_{\text{dev}}} \bar{E}_u$ ;
7   for  $t \leftarrow 2$  to  $n$  do
8      $\pi(\sigma_t) \leftarrow \arg \min_{u \notin \pi(\{\sigma_1, \dots, \sigma_{t-1}\})} \sum_{s < t} w_{\sigma_t \sigma_s} d(u, \pi(\sigma_s)) E_{u\pi(\sigma_s)}$ ; // greedy step
9   Build  $U(\boldsymbol{\gamma}^0, \boldsymbol{\beta}^0)$  on  $\pi$ : for  $k = 1, \dots, p$  apply  $e^{-i\gamma_k^0 H_C}$ , then  $R_{XX+YY}(\beta_k^0)$  on every mapped edge;
10  foreach  $\lambda \in \Lambda$  do
11    Insert identity/delay gates so that the effective noise of  $U$  is scaled by  $\lambda$ , using the device  $T_1, T_2$ ;
12    Execute  $S$  shots; record counts  $C_\lambda$  and  $\hat{E}_\lambda \leftarrow \langle H_C \rangle_{C_\lambda}$ ;
13   $\hat{E}_0 \leftarrow 2\hat{E}_1 - \hat{E}_2$ ; // extrapolate to  $\lambda = 0$ 
14  Compute QNRE from  $\hat{E}_0$  and the noise floor; update the Bayesian posterior over  $p$ ;
15 return  $\hat{E}_0, \mathbf{x}^* \leftarrow \arg \max_{\mathbf{x}} C_1(\mathbf{x})$  among feasible low-energy strings, QNRE;
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